DATABASE MANAGEMENT SYSTEMS · PROJECT PHASE 1

HOW TO WRITE THE CASE STUDY

Submit next week: the written **CASE STUDY only** – soft copy – online mode on or before 5th October 2026
Do NOT submit: the ER diagram, the relational schema, or SQL statements. Those are Phases 2, 3, 4 and 5.

Deliverable	Format	Mode	Due date
Case study document	PDF, typed, A4	Online	5th October 2026

1. What a Case Study Is

Your case study is the requirement specification of your system. Everything you build later — ER diagram, tables, keys, SQL — will be derived from this one document.

Note: It is not a description of your app, its screens, or its technology.

Write about this	Do not write about this
The real-world objects and the data stored about them	Screens, buttons, login pages, UI design
Who does what	Programming language, framework, hosting
Rules the organisation must follow	Advantages of DBMS over file systems
What information each transaction generates	Future scope, hardware requirements

2. The Seven-Step Writing Process

Follow the steps in order. Steps 1–3 are rough work; steps 4–7 produce the document you submit.

Step 1 — Choose the domain and fix its boundary

- Pick a real system with many interacting objects. Do not work on projects such as: calculator, to-do list, address book, simple marks register — these cannot produce a full ER model.
- Write one line each on: what the system does, who uses it, what is included, what is deliberately excluded.
- Fixing the boundary early prevents a case study that is half-finished.

Step 2 — Study two real-world reference systems

- Choose two established systems in the same domain.
- Note the information they clearly store: what you must fill in, what they show you, what appears on confirmation.
- Record both names with official websites, and list the specific features you took from each.
- **Use them as functional references only. Never copy their database design.**

Step 3 — Extract nouns and verbs from three real transactions

This is the single most reliable way to find every entity. Write out three core transactions of your system in plain sentences, then underline nouns (candidate entities and attributes) and verbs (candidate relationships).

- Then test each noun: does it have its own identity, its own attributes and its own lifecycle? If it only describes something else, it is an attribute, not an entity.

Step 4 — Document every entity

- Aim for atleast 8–12 entities. Fewer than 8 usually means you have missed the supporting entities.
- For each entity write: purpose, primary attribute, other attributes, and constraints.

- Give every entity a primary attribute. Avoid Name, City or Phone as the identifier unless strongly justified.
- Mark composite, multivalued and derived attributes where they occur naturally.

Step 5 — Document every relationship on both sides

- Name each relationship with a verb: places, operates, allots, issues, writes. Never "has" alone.
- Write one sentence for each direction, then the notation. Both minimum and maximum must appear on both sides.
- For every M:N relationship, state what data the association itself carries.

Step 6 — Write the business rules and back-check

- Write the rules as numbered plain-English statements.
- Back-check: take each cardinality you wrote in Step 5 and find the rule that produces it. If no rule produces it, the rule is missing or the cardinality is guessed.

Step 7 — Assemble, proofread, peer-reviewed and submit

- Arrange under the eight headings in Section 3 below, in points — never long paragraphs rather in **point form**.
- Get it peer-reviewed from friend, get the reviews.
- Run the checklist in Section 4 before submitting.

3. Required Structure of the Submission

#	Section	What it must contain
1	Project Title	One specific system name.
2	System Overview	Purpose, users, major activities, scope included and excluded (6–10 bullets).
3	Real-World References	Two systems with official websites, plus the features taken from each.
4	Entities and Attributes	Every entity: purpose, primary attribute, other attributes, constraints.
5	Relationships	Every relationship: name, both-side statement, notation, participation, relationship attributes, business rule.
6	Business Rules	Numbered rules from which every cardinality follows.
7	Special Constraints	Weak entities, mandatory rules, status values, restrictions.
8	ER Modelling Considerations	Your lists of 1:1, 1:M and M:N relationships; weak entities; composite, multivalued and derived attributes; relationship attributes.

4. Checklist — Tick Before You Submit

Any unticked row means the case study is incomplete and will lose marks.

#	Check	Done (Y/N)
1	Domain is realistic and complex enough for a full ER model.	
2	Scope stated — what is included and what is excluded.	
3	Two real-world reference systems named with official websites.	
4	Features studied from each reference are listed.	
5	No reference system's database design has been copied.	
6	System overview covers purpose, users, activities and stored information.	
7	8–12 entities identified, including supporting entities.	
8	Every entity has a stated purpose.	
9	Every entity has a primary attribute.	
10	Every entity has sufficient, meaningful attributes.	
11	Composite attributes marked where they occur.	
12	Multivalued attributes marked where they occur.	
13	Derived attributes marked where they occur.	
14	Weak entity identified and its owner entity stated (if applicable).	
15	Every relationship has a verb-based name.	
16	Every relationship is written from BOTH sides in sentences.	
17	Cardinality notation given on BOTH sides of every relationship.	
18	Participation (mandatory / optional) stated on BOTH sides.	
19	At least one M:N relationship identified.	
20	Data carried by each M:N relationship is described and justified.	
21	Business rules written as numbered plain-English statements.	
22	Every cardinality can be traced back to a business rule.	
23	Special constraints and status rules listed.	
24	ER modelling considerations section completed.	
25	Written in points and tables, not long paragraphs.	
26	No ER diagram, relational schema or SQL included.	
27	Peer-reviewed this document alone.	

What happens next

Phase 1 (this submission): Case Study.
Phase 2: ER Diagram derived from your case study.

> Phase 3: Mapping to the relational schema, with primary and foreign keys
>
> Phase 4: Normalisation.
>
> Phase 5: SQL implementation and queries.
>
> Every later phase is graded against this case study, so anything vague now becomes an error you have to carry forward.

Roll No.: _____

Student Name: _____

Section: _____

Domain Chosen: _____